Analytical Space Jacobian Elements of the 7R Mechanism

Automated Kinematics Engine

March 12, 2026

Introduction

This document presents the exact symbolic expressions for the space Jacobian matrix $J_s(q) \in \mathbb{R}^{6\times 7}$. Due to the extreme complexity of the trigonometric terms in distal links, each column J_i is decomposed into its 6 individual spatial components $J_{i,1} \dots J_{i,6}$.

Column 1 (J_1)

$$J_{1,1} = 0 \; J_{1,2} \quad = 0 \; J_{1,3} = 1 \; J_{1,4} \quad = 0 \; J_{1,5} = 0 \; J_{1,6} \quad = 0$$

Column 2 (J_2)

$$J_{2,1} = 0 \; J_{2,2} \quad = 0 \; J_{2,3} = 1 \; J_{2,4} \quad = 0 \; J_{2,5} = -l \; J_{2,6} \quad = 0$$

Column 3 (J_3)

$$J_{3,1} = \sin(\mathrm{q1}) \; J_{3,2} \quad = -\cos(\mathrm{q1}) \; J_{3,3} = 0 \; J_{3,4} \quad = 0 \; J_{3,5} = 0 \; J_{3,6} \quad = -l(\cos(\mathrm{q1}) + 2)$$

Column 4 (J_4)

$$J_{4,1} = \cos(\mathrm{q1})\sin(\mathrm{q2}) \; J_{4,2} \quad = \sin(\mathrm{q1})\sin(\mathrm{q2}) \; J_{4,3} = -\cos(\mathrm{q2}) \; J_{4,4} \quad = \frac{1}{2} l \cos(\mathrm{q2})(5\cos(\mathrm{q1}) - 4\sin(\mathrm{q1})) \; J_{4,5} = \frac{1}{2} l \cos(\mathrm{q2})(5\sin(\mathrm{q}$$

Column 5 (J_5)

$$J_{5,1} = \sin(\mathrm{q1})\cos(\mathrm{q3}) - \cos(\mathrm{q1})\cos(\mathrm{q2})\sin(\mathrm{q3}) \; J_{5,2} \quad = -\sin(\mathrm{q1})\cos(\mathrm{q2})\sin(\mathrm{q3}) - \cos(\mathrm{q1})\cos(\mathrm{q3}) \; J_{5,3} = -\sin(\mathrm{q2})\sin(\mathrm{q3}) \; J_{5,4}$$

Column 6 (J_6)

$$J_{6,1} = -\cos(\mathrm{q1})(\cos(\mathrm{q2})\cos(\mathrm{q3})\sin(\mathrm{q4}) + \sin(\mathrm{q2})\cos(\mathrm{q4})) - \sin(\mathrm{q1})\sin(\mathrm{q3})\sin(\mathrm{q4}) \; J_{6,2} \quad = \sin(\mathrm{q4})(\cos(\mathrm{q1})\sin(\mathrm{q3}) - \sin(\mathrm{q1})\cos($$

Column 7 (J_7)

$$J_{7,1} = -\cos(\mathrm{q3})(\cos(\mathrm{q1})\cos(\mathrm{q2})\cos(\mathrm{q4})\sin(\mathrm{q5}) + \sin(\mathrm{q1})\cos(\mathrm{q5})) + \cos(\mathrm{q1})(\cos(\mathrm{q2})\sin(\mathrm{q3})\cos(\mathrm{q5}) + \sin(\mathrm{q2})\sin(\mathrm{q4})\sin(\mathrm{q5})) + s$$